Benchmarking Spatial and Affordance Reasoning in Vision-Language Models for Robotics

Maurice Behanzin Céline Sonnenschein Nina Gruteser Mathilda Lee
Supervisors: Jovana Videnovic Jelena Trisovic René Zurbrügg

Abstract

Vision-language models (VLMs) are increasingly used as perception and reasoning modules for robotics, but aggregate visual-question-answering scores do not show whether a model understands the spatial and physical constraints that matter for manipulation. This report builds a targeted evaluation layer on top of Robo2VLM, categorizing its test questions into spatial reasoning, affordance understanding, or neither, then evaluating open VLMs under zero-shot, chain-of-thought, subcategory, and visual-grounding controls on only the spatial reasoning and affordance understanding questions. This study aims to benchmark and model failure modes that normal evaluation can hide. Across the model runs, spatial reasoning remains a weakness, affordance understanding scores are dominated by object-blockage questions, chain-of-thought is not reliably beneficial, additional depth information does not improve accuracy, and sanity probes reveal that accuracy alone is not sufficient evidence for robotics reasoning.

1. Introduction

Robotic manipulation depends on visual reasoning that is more specific than generic image understanding. A robot must infer which object is reachable, whether a grasp is stable, which point is closer than the other in 3D, and which direction to move the end-effector next to reach a desired pose. Recent vision-language-action systems show that pretrained VLMs can transfer semantic knowledge into robot planning and control [2, 6, 11, 14]. At the same time, spatial reasoning and grounded affordance understanding remain known weak points for VLMs, motivating targeted benchmarks to discover which, if any, VLMs possess these qualities. [3, 5, 13].

Robo2VLM is a recent robotics vision-question-answer (VQA) benchmark generated from real robot trajectories, with questions rooted in robot state, end-effector pose, force, task phase, and scene information [4]. It is valuable because it transforms manipulation trajectories into multiple-choice VQA examples. However, a broad Robo2VLM score can mix several qualities: geometric reasoning, goal-state matching, task completion recognition, obstacle reasoning, and answer-template priors. This study aims to find if current VLMs actually solve the embodied reasoning problem when Robo2VLM questions are filtered for spatial reasoning and affordance understanding.

Contributions. This project contributes an evaluation layer over Robo2VLM that reveals critical flaws in aggregate benchmark interpretation and in the tested VLMs:

- a curated Robo2VLM test split into spatial reasoning, affordance understanding, and neither, with subcategories for depth/direction/correspondence and object-blockage/grasp-stability questions;
- an automated testing pipeline that grades on explicit multiple-choice letter selection, testing models with zero-shot and chain-of-thought prompts, RGB and RGB-depth image modes, and altered images;
- a comparison of Qwen2.5-VL-3B, DeepSeek-VL2-tiny, and additional compact models on spatial and affordance subsets;
- sanity probes that test whether explanations remain tied to the image under correct-answer, wrong-answer, blank-image, and shuffled-image conditions.

The main finding is that spatial reasoning scores remain low, affordance understanding scores are stronger but largely driven by object-blockage questions, and chain-of-thought does not reliably improve performance.

2. Related Work

2.1. Robotics VLMs and trajectory-scale data

Embodied VLMs and VLAs such as PaLM-E, RT-2, and OpenVLA integrate visual-language representations with robot data to improve planning or action generation [2, 6, 11]. These models are enabled by large manipulation datasets such as Open X-Embodiment and DROID, which broaden the diversity of robots, tasks, and scenes available for learning [10, 14]. Robo2VLM takes a complementary direction: it uses robot trajectories to generate VQA examples for evaluating and improving VLMs rather than directly training a control policy [4].

2.2. Visual and spatial reasoning benchmarks

Classic VQA and compositional reasoning datasets showed that aggregate accuracy can hide question biases and reasoning failures [8, 9]. Spatial reasoning benchmarks such as VSR, SpatialVLM, and SpatialRGPT further show that VLMs struggle with relative position, orientation, distance, and 3D-aware region reasoning [3, 5, 13]. Our benchmark slice follows this diagnostic spirit, but evaluates robotics images and manipulation-specific questions rather than web images or synthetic scenes.

2.3. Grounding and hallucination checks

Multimodal benchmarks such as MME evaluate perception and cognition across a broad range of tasks [7], while hallucination-focused benchmarks such as POPE examine whether model outputs remain grounded in image content [12]. Inspired by these diagnostic evaluations, we introduce a set of grounding probes for robotics VQA. These probes assess whether a model can distinguish supported from unsupported claims by evaluating its responses under blank-image, shuffled-image, and deliberately incorrect-answer controls.

3. Benchmark Design

3.1. Scoring methodology

Let a Robo2VLM test item be (x_i, q_i, C_i, y_i), where x_i is an RGB robot image, q_i is a natural-language question, $C_i = \{c_{iA}, \dots, c_{iE}\}$ is a multiple-choice set, and y_i is the correct option letter. We curate each item into a label

$$\ell_i \in \{\text{spatial}, \text{affordance}, \text{neither}\}, \tag{1}$$

then assign deterministic subcategories from question text. Spatial questions are grouped into distance and depth estimation, directional orientation, and cross-view correspondence questions. Affordance questions are grouped into object-blockage and grasp-stability questions. Goal-state matching questions are excluded from the spatial subset because they test task-state recognition rather than spatial reasoning or affordance understanding.

For a model M, prompt mode p, and optional image transform a, for example adding depth information, the evaluator forms the prompt

$$z_i = p(q_i, C_i), \qquad r_i = M(a(x_i), z_i), \tag{2}$$

extracts the last standalone letter in $\{A, B, C, D, E\}$ from r_i, and computes

$$\text{Acc}(M, p, S) = \frac{1}{|S|} \sum_{i \in S} \mathbf{1}[\hat{y}_i = y_i]. \tag{3}$$

The two prompt modes are zero-shot (ZS), which requests only the option letter, and chain-of-thought (CoT), which asks for step-by-step reasoning followed by the final letter. Since CoT responses may contain several option letters in the reasoning text, the final standalone letter controls scoring.

Label	Subcategory	Count
Spatial	distance/depth	1,283
Spatial	direction arrows	798
Spatial	other	3
Affordance	object blockage	543
Affordance	grasp stability	299
Neither	–	3,750

Table 1. Curated Robo2VLM-1 test split used to construct the evaluation subsets.

3.2. Curation and sampling

To enable a more focused evaluation of spatial reasoning and affordance understanding, we curated the Robo2VLM test split into three high-level categories: *spatial*, *affordance*, and *neither*. An LLM-assisted annotation process was used to categorize each question according to the primary type of reasoning required to answer it. The resulting curated test split contains 6,676 question–image pairs, including 2,084 spatial questions, 842 affordance questions, and 3,750 questions that do not fall into either category.

To provide additional insight into the composition of the benchmark, we further assigned deterministic subcategory labels based on the question text. Spatial questions were divided into distance/depth estimation, directional orientation, and other spatial reasoning categories, while affordance questions were divided into object-blockage and grasp-stability categories. The distribution of questions across all categories and subcategories is summarized in Tab. 1. Examples can be seen in Fig. 1.

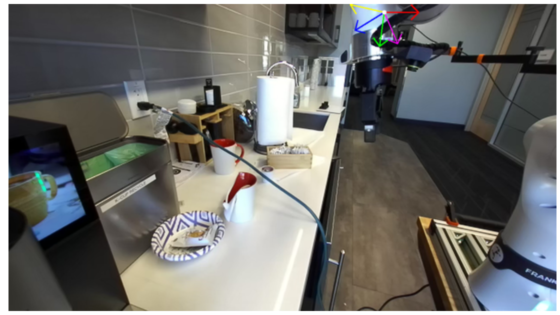
3.3. Sanity probes

The scored runs are complemented by raw-generation probes over 20 spatial CoT examples per model. For each example, we ask the model to justify the correct answer on the original image, assess a deliberately wrong answer, assess the correct answer on a blank image, assess the correct answer on a shuffled mismatched image, and describe overlays without selecting an answer. A phrase-based heuristic labels each response as supporting, rejecting/insufficient, or unclear. These labels are for triage, not a substitute for manual grounding evaluation.

3.4. Human performance evaluation

To complement the automated model evaluation, a human performance study was conducted on a subset of the benchmark. This analysis was motivated by the observation

(a) Object blockage/reachability, sample 20. Question: *Is there any obstacle blocking the robot from reaching chip bag?* Choices: *A. Partially reachable B. Cannot be determined C. No D. Yes*



(b) Directional orientation, sample 368. Question: *The robot task is to pick up the object. Which colored arrow correctly shows the direction the robot will move next?* Choices: *A. Red B. Green C. Blue D. Yellow E. Purple*

(c) Depth/distance ordering, sample 407. Question: *In the image from ext1, which colored point is FARTHEST from the camera?* Choices: *A. Purple B. Red C. Blue D. Green E. Yellow*

(d) Cross-view correspondence, sample 408. Question: *In the left image (ext1 camera), a red dot is marked. Which point is the closest point in the right image (ext2 camera) corresponding to the same 3D location?* Choices: *A. E B. A C. B D. D E. C*

Figure 1. Actual Robo2VLM examples used to ground the benchmark categories.

that the evaluated vision-language models exhibited only marginal performance differences across settings, prompting a closer inspection of the dataset itself. A manual review revealed that a subset of samples contains ambiguities, incomplete visual information, or inherently under-specified questions, as can be seen in Fig. 2, which can make them challenging even for human annotators.

To quantify the impact of such issues, a random subset of 50 question–image pairs was selected from the curated dataset and presented to a group of human participants. The participants were asked to answer each question based solely on the provided image and task description, without access to external information or prior context.

Figure 2. Ambiguous data sample example: missing colored arrows. Q: The robot task is to put the marker in the cup. Which colored arrow correctly shows the direction the robot will move next? A:Yellow, B: Green, C: Red, D: Purple, E: Blue

4. Experimental Setup

The main raw-result matrix evaluates Qwen2.5-VL-3B-Instruct and DeepSeek-VL2-tiny on the spatial and affordance subsets, with zero-shot and CoT prompts. All audited runs use Robo2VLM-1 test images, RGB input, temperature 0.0, batch size 1, tensor parallel size 1, and 200 examples per subset. Zero-shot generations are capped at 128 tokens, while CoT runs allow longer responses. Additional poster-reported compact baselines are included in Tab. 4.

The evaluated tasks are five-way multiple choice, so uniform chance is 20%. Response time is measured per generated answer. For prompt comparisons, we pair zero-shot and CoT results by question ID and count cases where zero-shot is correct but CoT is wrong, and vice versa.

To evaluate whether depth information improved spatial reasoning, we generated monocular depth maps for images in the Robo2VLM-1 test split using the pretrained Depth Anything V2 model (Small) [16]. For each test example, the inferred depth map was converted to a three-channel format and horizontally concatenated to the right of the original RGB image. This side-by-side strategy allows us to feed spatial priors directly into the VLMs as a single standard image without requiring architectural modifications. We also included relevant information in the prompt, explicitly telling the VLM that the left side had the original image and the right side had the depth image as in Fig. 1c.

5. Results

5.1. Qwen and DeepSeek evaluations

Tab. 2 shows the main Qwen and DeepSeek matrix. Spatial reasoning is essentially unsolved in this setting: the best audited spatial result is 22.0%, only two percentage points above five-way chance. Affordance results are stronger, but the subcategory analysis in Tab. 3 shows that this strength is mostly due to object-blockage questions rather than stable grasp reasoning.

Model	Spatial ZS	Spatial CoT
Qwen2.5-VL-3B	40/200, 20.0% (0.18s)	39/200, 19.5% (4.90s)
DeepSeek-VL2-tiny	39/200, 19.5% (0.17s)	44/200, 22.0% (0.74s)
Model	**Affordance ZS**	**Affordance CoT**
Qwen2.5-VL-3B	131/200, 65.5% (0.46s)	128/200, 64.0% (3.40s)
DeepSeek-VL2-tiny	144/200, 72.0% (0.14s)	43/200, 21.5% (0.27s)

Table 2. Main audited 200-example results. Cells report correct/total, accuracy, and average response time.

5.2. Effect of Prompting Strategy: zero-shot vs. CoT

CoT changes behavior but does not reliably improve accuracy. For Qwen2.5-VL-3B, CoT slightly hurts both spatial and affordance performance: -0.5 percentage points on spatial and -1.5 on affordance. DeepSeek-VL2-tiny gains only $+2.5$ points on spatial, but collapses on affordance from 72.0% zero-shot to 21.5% CoT. In the paired DeepSeek affordance comparison, 114 examples are correct in zero-shot but wrong under CoT, while only 13 move from wrong to correct. A similar trend can be seen in Tab. 4 while evaluating additional models. This is the clearest evidence that reasoning traces can introduce systematic errors rather than merely exposing latent reasoning.

Model	Prompt	Direction	Dist/Depth	Cross-view
Qwen2.5-VL-3B	ZS	8/76, 10.5%	24/91, 26.4%	8/33, 24.2%
Qwen2.5-VL-3B	CoT	12/76, 15.8%	21/91, 23.1%	6/33, 18.2%
DeepSeek-VL2-tiny	ZS	14/76, 18.4%	17/91, 18.7%	8/33, 24.2%
DeepSeek-VL2-tiny	CoT	15/76, 19.7%	21/91, 23.1%	8/33, 24.2%
Model	**Prompt**	**Obj. Blockage**	**Grasp Stability**	
Qwen2.5-VL-3B	ZS	105/132, 79.5%	26/68, 38.2%	
Qwen2.5-VL-3B	CoT	107/132, 81.1%	21/68, 30.9%	
DeepSeek-VL2-tiny	ZS	112/132, 84.8%	32/68, 47.1%	
DeepSeek-VL2-tiny	CoT	18/132, 13.6%	25/68, 36.8%	

Table 3. Subcategory breakdown for the audited 200-example spatial and affordance slices. Spatial performance is weak across all subcategories. Affordance accuracy is dominated by object-blockage questions; grasp stability remains much harder.

Model	Spatial ZS	Aff. ZS	Spatial CoT	Aff. CoT
SmolVLM-2.2B	20%	23%	66%	47%
Phi-4-multimodal	12%	19%	40%	44%

Table 4. Additional compact-model results.

5.3. Effect of Depth Information

Contrary to expectations, depth information did not show any improvement in model accuracy. Qwen2.5-VL-3B was benchmarked on a dataset of 2084 spatial reasoning examples, which were augmented by concatenating the depth image to the side of the original image. Evaluated with zero-shot prompting, accuracy actually decreased slightly to 18.3% on RGB-depth images.

5.4. Sanity probes results

Tab. 5 summarizes the raw image-understanding sanity probes. Both audited models reject blank images reliably, so the image channel is not entirely ignored. The more informative controls are wrong answers and shuffled images. Qwen supports a deliberately wrong answer in 16/20 cases and rejects only 3/20 shuffled-image controls. DeepSeek rejects all shuffled controls, but its correct-answer explanations are usually classified as unclear. These probes show why answer accuracy alone is not sufficient evidence for grounded robotics reasoning.

5.5. Human baseline

The resulting human performance provides an additional reference point for interpreting model results and for assessing benchmark validity. In particular, it helps determine whether incorrect model predictions stem from limitations in model reasoning or from ambiguities inherent in the benchmark itself. Tab. 6 indicates that VLMs outperform human participants in overall affordance understanding, with the most pronounced advantage observed in the grasp stability subcategory. In contrast, humans demonstrate stronger performance in spatial reasoning tasks. However, it is noteworthy that performance in the direction subcategory remains close to chance level (five-way guessing), suggesting that both humans and models struggle with this aspect of spatial reasoning using the Robo2VLM dataset.

Spatial	Direction	Distance/Depth	Cross-view
Total: 43.3%	26.9%	52.5%	42.0%
Affordance	**Object Blockage**	**Grasp Stability**	
Total: 45.0%	63.3%	22.7%	

Table 6. Human performance using a randomized 50 data-samples subset.

Probe	Expected behavior	Qwen2.5-VL-3B	DeepSeek-VL2-tiny
Correct answer	Support correct answer on original image	12 supports, 8 unclear	**20 unclear**
Wrong answer	Reject deliberately wrong answer	**16 supports**, 4 unclear	3 rejects, 17 unclear
Blank image	Reject image as insufficient	20 rejects	20 rejects
Shuffled image	Reject mismatched image	**3 rejects**, 17 unclear	**20 rejects**
Overlay description	Describe visual overlays without answering	20 unclear	20 unclear

Table 5. Heuristic labels for 20 examples per probe type. Bold values mark behavior most relevant to the grounding claim: unclear correct-answer support, wrong-answer rationalization, and mismatched-image rejection or failure to reject.

6. Discussion

Aggregate benchmark scores can be misleading. The affordance subset appears substantially easier than spatial reasoning in the headline table, but Tab. 3 shows that this is not uniform affordance competence. Object-blockage questions drive most of the score, while grasp stability remains below 50% for every audited run. This matters for robotics: a robot that can answer obstacle questions from template cues may still fail to judge whether a grasp is physically stable.

Spatial reasoning remains a critical weakness. The audited spatial results cluster around chance despite evaluating models with strong general visual-language capabilities [1, 15]. Directional orientation, distance/depth ordering, and cross-view correspondence questions all remain weak. The visual categories in Fig. 1 show why these questions are nontrivial: colored overlays must be interpreted relative to camera geometry, robot pose, and multi-view alignment.

CoT is an ablation, not a default improvement. CoT is often treated as a low-cost way to improve reasoning. Here, it is at best mixed. Qwen changes many individual decisions without improving aggregate accuracy. DeepSeek gains a small amount on spatial questions but fails dramatically on affordance CoT, especially object blockage. The example in Fig. 3 shows one concrete failure mode: target/obstacle confusion. This is not merely random noise; it matches the aggregate fall from 84.8% to 13.6% on object blockage/reachability.

Grounded explanations need separate tests. The sanity probes in Tab. 5 reveal a distinction between answering and grounding. Qwen correctly rejects blank images, yet it also supports deliberately wrong answers and often fails to reject shuffled images. DeepSeek handles shuffled images better, but its explanations are often not phrased as explicit support for the known correct answer. A benchmark that reports only multiple-choice accuracy misses these behaviors.

Limitations. The main raw-result matrix uses 200 examples per subset and file-order sampling. The sanity probe heuristic is coarse and should be followed by manual review before making fine-grained claims about explanation quality. The benchmark also inherits Robo2VLM's multiple-choice format, template distribution, and generated-answer assumptions. These limitations do not weaken the central diagnostic result: targeted slicing and controls expose failures that aggregate robotics VQA scores can hide.

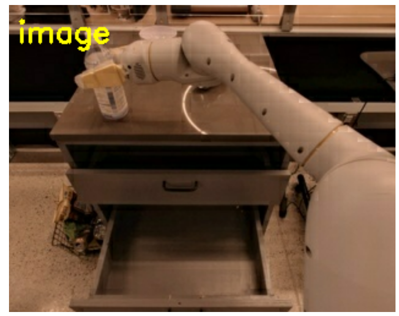


Figure 3. Reasoning Collapse: Enabling CoT reasoning leads to the model *DeepSeek VL2 tiny* treating the target object itself, the bottle, as an obstacle.

The scope of the evaluation was constrained by available computational resources, limiting the study to relatively small VLMs. Since model scaling has been shown to improve performance across a range of reasoning tasks, it remains an open question whether larger models would exhibit stronger spatial and affordance reasoning abilities. Exploring this relationship constitutes an important avenue for future research.

Data quality limitations. A key finding from the human evaluation is that performance is not consistently high even for human participants. Several samples were identified as overly ambiguous, under-specified, or otherwise too difficult to be reliably solved based on the provided image and question alone, as presented in Fig. 2. This indicates that a non-negligible portion of the dataset does not constitute well-posed evaluation cases for spatial and affordance reasoning.

As a consequence, the benchmark in its current form cannot be directly used as-is for reliable model evaluation. Instead, it requires additional manual filtering and curation to isolate representative and well-defined task instances. Such handpicking of high-quality samples is necessary to ensure that the final benchmark accurately reflects the intended reasoning capabilities and provides a meaningful basis for comparing model performance.

7. Conclusion

This project adds a targeted diagnostic layer to Robo2VLM for evaluating spatial reasoning and affordance understanding in robotics VQA. The results show that current VLMs can appear competent on aggregate affordance questions while still failing grasp stability, depth, direction, cross-view correspondence, and grounding checks. The report therefore supports a conservative recommendation: robotics VLM benchmarks should report ability-specific slices, prompt-mode ablations, and image-grounding controls rather than relying on a single VQA accuracy number.

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